Recommended Assessment

Passive Infrared

Exploring Passive Infrared Sensors

1. Do you expect the field of view will be affected by small/large temperature changes? Will the distance of the object also affect the field of view?

2. What do you notice about the measured field of view compared to the field of view stated in PIR-PL-Q873-02.pdf?

3. Based on the tests performed, are Passive Infrared Sensors good for measuring distance, motion, both? Why?

4. What are some system design considerations if you were to use a Passive Infrared sensor in a mechatronics application?

Real Life Scenarios

5. You are building a temperature-controlled room for testing automotive parts. A safety system is used to ensure a person does not enter the room while in operation. Would a passive infrared sensor be useful for this application? What steps would you need to consider if you were going to install a passive infrared motion detector?

6. Would it be possible to bypass a motion detection system which relied on just a passive infrared sensor? How would you augment this system to ensure redundancy?

Passive Infrared Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	